Ziyi Wang

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Education

Northeastern University, Boston, MA, USA — Sep. 2024 – Present
Ph.D. student in Computer Science. Advisor: Dakuo Wang

Tsinghua University, Beijing, China — Aug. 2021 – Jun. 2024
Master's in Electronic and Information Engineering. Advisor: Ji Wu

Tsinghua University, Beijing, China — Aug. 2017 – Jun. 2021
Bachelor of Engineering in Electronic Information Science and Technology

Research Interests

My research generally focuses on LLM agents, with emphasis on (1) synthesizing verifiable data for tool-use training, (2) post-training with SFT and RL to improve agent capabilities, and (3) personalized simulation of human behavior with LLMs.

Publications

- **Trajectory2Task: Training Robust Tool-Calling Agents with Synthesized Yet Verifiable Data for Complex User Intents**.
  Ziyi Wang, Yuxuan Lu, Yimeng Zhang, Ziwei Dong, Jing Huang, Jiri Gesi, Xianfeng Tang, Chen Luo, Yisi Sang, Hanqing Lu, Manling Li, Dakuo Wang
  (Accepted to ACL 2026)
- **OPeRA: A Dataset of Observation, Persona, Rationale, and Action for Evaluating LLMs on Human Online Shopping Behavior Simulation**.
  Ziyi Wang, Yuxuan Lu, Wenbo Li, Amirali Amini, Bo Sun, Yakov Bart, Weimin Lyu, Jiri Gesi, Tian Wang, Jing Huang, Yu Su, Upol Ehsan, Malihe Alikhani, Toby Jia-Jun Li, Lydia Chilton, Dakuo Wang.
  (Accepted to ACL 2026, NeurIPS Workshop 2025 on MTI-LLM and SEA)
- **SENTINEL: Failure-Driven Reinforcement Learning for Training Tool-Using Language Model Agents**.
  Ziyi Wang, Yuxuan Lu, Yimeng Zhang, Qun Liu, Chen Luo, Jiri Gesi, Hanqing Lu, Yisi Sang, Manling Li, Jing Huang, Dakuo Wang.
  (Preprint)
- **Firefly: Illuminating Large-Scale Verified Tool-Call Data Generation from Real APIs**.
  Yuxuan Lu, **Ziyi Wang**, Yingzhou Lu, Yisi Sang, Jiri Gesi, Xianfeng Tang, Yimeng Zhang, Zhenwei Dai, Hui Liu, Hanqing Lu, Chen Luo, Qi He, Benoit Dumoulin, Jing Huang, Dakuo Wang.
  (Preprint)
- **Customer-R1: Personalized Simulation of Human Behaviors via RL-based LLM Agent in Online Shopping**.
  Ziyi Wang, Baber Khalid, Houyu Zhang, Jiri Gesi, Yuxuan Lu, Yimeng Zhang, Tian Wang, Ruochen Jiao, Alejandro Mottini, Jing Xu, Yufan Guo, Qingjun Cui, Dakuo Wang.
  (Accepted to NeurIPS Workshop 2025 on MTI-LLM)
- **Shop-R1: Rewarding LLMs to Simulate Human Behavior in Online Shopping via Reinforcement Learning**.
  Yimeng Zhang, Tian Wang, Jiri Gesi, **Ziyi Wang**, Yuxuan Lu, Jiacheng Lin, Sinong Zhan, Vianne Gao, Ruochen Jiao, Junze Liu, Kun Qian, Yuxin Tang, Ran Xue, Houyu Zhang, Qingjun Cui, Yufan Guo, Dakuo Wang.
  (Accepted to ICLR 2026)

Internship

Applied Scientist Intern @ Amazon.Com, Inc, Palo Alto, USA — Sep. 2025 – Present

Working on training tool-calling agents with synthetic data.

Applied Scientist Intern @ Amazon.Com, Inc, Seattle, USA — May. 2025 – Aug. 2025

Worked on personalized simulation of human behaviors via RL-based LLM agent.

Research Intern @ Xiaomi Corp., Beijing, China — Aug. 2023 – Nov. 2023

Worked on autonomous driving planning algorithms using multimodal large model.

Research Intern @ Megvii, Beijing, China — Aug. 2021 – Jan. 2022

Worked on robustness of face recognition models.

Research Experiences

SENTINEL: Failure-Driven RL for Tool-Using Agents — Feb. 2026 – May. 2026

Amazon | *Applied Scientist Intern*

- Proposed a failure-driven reinforcement learning framework that turns the Solver's rollout failures into targeted training tasks via a Controller–Proposer–Solver loop.
- Addressed the task distribution mismatch problem in RL training: as the policy evolves, fixed tasks become uninformative; SENTINEL adaptively generates tasks targeting the model's current weaknesses without human annotation.
- Improved Pass^1 from 66.4 to 74.9 on Tau2-Bench Retail with Qwen3-4B, outperforming RL on general synthetic tasks across Pass^k metrics.

Firefly: Verified Tool-Call Data Synthesis from MCP Servers — Dec. 2025 – May. 2026

Northeastern University | *PhD Student*

- Proposed a scalable framework for generating verifiable tool-calling data from real MCP servers via graph-guided exploration and backward task synthesis.
- Constructed a dataset of 5,144 verified multi-turn tasks spanning 240 MCP servers and 993 tools.
- Demonstrated improved tool-calling performance on Tau2-Bench, MCPMark, and MCP-Atlas, with a 4B model achieving results competitive with Claude Sonnet 4.6.

Trajectory2Task: Learning Tool Use in Real-World Settings — Sep. 2025 – Jan. 2026

Amazon | *Applied Scientist Intern*

- Proposed a systematic study of tool-calling agents under ambiguous, changing, and infeasible user goals, analyzing multi-turn decision-making in realistic tool-use settings.
- Developed a general, verifiable data synthesis pipeline that generates complex multi-turn tool-use tasks and trajectories, and built a comprehensive benchmark revealing key LLM failure modes.
- Proposed a trajectory-based SFT training method that teaches agents when to ask, how to adapt, and how to act, improving tool-calling performance and transferring to unseen domains.

Customer-R1: Personalized Simulation of Human Behavior with RL — May. 2025 – Aug. 2025

Amazon | *Applied Scientist Intern*

- Developed an RL-based algorithm for step-by-step user behavior simulation in online shopping, optimizing both action generation and reasoning alignment.
- Leveraged real human behavior data from the OPeRA dataset with custom reward signals to improve fidelity of simulated behaviors.
- Compared against supervised fine-tuning and prompting baselines; achieved better alignment with authentic human decision-making processes.

OPeRA: User Online Shopping Behavior Simulation Benchmark Oct. 2024 – May. 2025

Northeastern University | PhD Student

- Designed and implemented a Chrome extension to collect detailed user behavior data on Amazon.com, including click, scroll, and navigation patterns.
- Built a comprehensive dataset capturing user shopping behavior, environmental context, final purchasing decisions, rationales, and persona attributes.
- Developed evaluation tasks to benchmark LLM agents and support research on human behavior simulation.

Prompt Based Automatic ICD Coding with Knowledge Injection Jun. 2023 – Apr. 2024

Tsinghua University | Research Assistant

- Developed a unified knowledge-injection approach to enhance pre-trained models in specialized domains, enabling extensible incorporation of new knowledge.
- Introduced a GNN pre-training method tailored for hierarchical knowledge data.
- Designed an ICD coding algorithm using GNN pre-training and multimodal training strategies with prompt-tuning; achieved SOTA on MIMIC-III-50, especially in few-shot settings.

Autonomous Driving Planning Algorithms Jul. 2023 – Nov. 2023

Xiaomi | Research Intern

- Contributed to an end-to-end autonomous driving decision and planning model integrating multi-sensor and high-level inputs.
- Designed methods using navigation information to improve holistic route understanding.
- Built data interfaces and optimized preprocessing; established evaluation on NuPlan and CARLA Leaderboard, conducting closed-loop tests at 2Hz on CARLA.

♡ Honors and Awards

- Northeastern University Khoury Distinguished Fellowship — Fall 2024
- Best Performance Prize, Intelligent Robot Obstacle Avoidance Competition — Dec. 2019
- Excellence Award, Tsinghua Electronic Design Competition — Dec. 2018
- Second Prize, Tsinghua Freshman Scholarship — Oct. 2017